

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR
MID-Autumn Semester 2025-26

Date of Exam: 09.09.2025
 Subject Number MA60225
 Subject Name: Financial Mathematics

Session(FN/AN)
 Max. Marks: 30

Duration: 2 hrs
 Department: Mathematics
 No. of students: 126

- Instructions:** (i) Use of calculator and Statistical tables is allowed.
 (ii) All answers of numerical questions must be in at least two decimal places.
 (iii) All the notations are standard. If any data/statement is missing, identify it on your answer script.
 (iv) Answer All questions. Marks are indicated at the end of each part of question.
 (v) All parts of a question **Must Be** answered at **One Place**.

For this question (Question 1 only) write only answers on the first page of your answer script in the given tabular form. Detail working may be carried out on other pages. [1×6]

Q. No. 1	a	b	c	d	e	f
Answer						

- (a) Suppose we have the option of buying, at some time in the future, one unit of a stock at a fixed price Rs. 100, independent of its current market price. The current market price of the stock is taken to be 0, and we suppose that it changes in accordance with a Brownian motion process having a drift coefficient -1. When, if ever, should we exercise our option?
- (b) Suppose that $\{Y(t), t \geq 0\}$, is a geometric Brownian motion with drift parameter $\mu = 0.01$ and volatility parameter $\sigma = 0.2$. If $Y(0) = 10$. Then, find the value of $E[Y(100)]$.
- (c) The present price of a stock is 50. The price at time 1 will be either 25 or 100. An option to purchase y shares of the stock at time 1 for the (present value) price Ky costs cy . If $K = 85$, for which values of c an arbitrage opportunity occurs?
- (d) Let $\{W_t, t \geq 0\}$ be a Wiener process, then find the distribution of $W_1 - W_2 + 3W_4$.
- (e) Suppose $\{W(t), t \geq 0\}$ be a Wiener process. Let $c > 0$. Then, $V(t) = \frac{1}{c}W(c^2t)$ is also a Wiener process. (T/F)
- (f) Let $\{W_t, t \geq 0\}$ be a Wiener process, then $|W_t|^2 - t$ is NOT a martingale (MG). (T/F)

2/ Write short note on following:

[2×3]

- (a) American and European call options.
 (b) Derive Ito's formula.
 (c) Gamblers ruin problem.

3. (a) For dynamic programming technique in Markov decision processes, derive the optimality equation using the total probability formula. [3]

- (b) At each play of the game, a gambler can bet any non-negative amount up to his present fortune and will either win or lose that amount with probabilities $2/3$ and $1/3$ respectively. The gambler is allowed to make $T = 3$ bets, and his objective is to maximize the expected Bernoulli utility of his final outcome. What is the optimal strategy? Obtain it using optimality equation derived in part(a). [3]

4. (a) Consider the following stochastic differential equation (SDE) with respect to a Wiener process W_t as

$$dX_t = dt + 2\sqrt{X_t} dW_t, \quad X_0 = 0.$$

Show that it has solution $X_t = W_t^2$. Also find $E(X_t)$ and $\text{Var}(X_t)$. [3]

- (b) Let

$$X_t := e^{t/2} \cos(W_t), \quad t \geq 0,$$

where $\{W_t\}_{t \geq 0}$ be a Wiener process.

- (i) Apply Itô's formula to compute the stochastic differential dX_t .

- (ii) Using the result part (i) or otherwise: Evaluate $E(\cos(W_t))$. [3]

5. (a) Let τ_a denote the hitting time of level $a > 0$ for an Wiener process. Then find the cdf and pdf of τ_a . [3]

- (b) Let $W(t)$ be a Wiener process and $M(t) = \max_{0 \leq u \leq t} W(u)$. Find

$$P(1 < W(4) < 3, 2 < M(4) < 3).$$

[3]
